How-To: Switchable Execution Backends and Sandboxing

ragent can run a session's tool invocations in one of four **execution backends**: on the host (`local`, the default), inside a container sandbox (`docker` or `podman`), or against a second ragent server reached over its REST+SSE API (`remote`). Selecting a backend changes *where* a tool runs - it never changes the agent loop, the LLM providers, the tool registry, the permission system, or the event bus (spec `openhands` FR-001, assumption A1).

This guide covers four things:

- The `/backend` TUI command that lists, inspects, and switches the active backend.
- How the switchable backend subsystem works end to end - the trait, resolution, the registry, the per-session override, and the "never fall back to host" rule.
- The configuration that declares backends.
- A complete worked example: building an image, storing a credential, and running a session inside a Podman (or Docker) sandbox.

> **Scope.** This guide covers backend selection and sandboxing. For the static permission pipeline and the 7-layer bash model see `permissions.md`. For the config key reference see `config.md` section 7.36. For the `/backend` command quick reference see `slashcommands/backend.md`. For the requirements see `specs/openhands/SPEC.md` FR-001 through FR-037.

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1. Concepts

Term	Meaning
Execution backend	The layer that decides where one tool call executes. Selected by the `execution_backend` config key plus optional `backends` entries.
Kind	One of `local`, `docker`, `podman`, `remote` (`ExecutionBackendKind`, `crates/ragent-config/src/config.rs`).
Descriptor	A `BackendConfig` entry: a stable `id`, a display `name`, a `kind`, kind-specific connection fields (`image`, `workspace`, `url`, `api_key`), and a list of credential *names*.
Registry entry	The read model over a descriptor: id, name, kind, a secret-free `ConnectionDescriptor`, and a health state (FR-004).
Sandbox	The long-lived container a `docker`/`podman` backend provisions for a session and dispatches every tool call into.

The important invariants (FR-001, FR-002, FR-037): backend selection is confined to the *tool-execution leaf*. The 7-layer bash validator, the file-path containment guard, the permission system, and the always-allowed codeindex hardwiring all run **before** any backend dispatch and are unchanged by the choice.

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2. The `/backend` Command

`/backend` is the TUI surface for the execution backend (FR-008). It is implemented in `crates/ragent-tui/src/app/slash.` (`handle_backend_command`, dispatch arm `"backend"`) and `crates/ragent-tui/src/app/backend_panel.rs`.
It is TUI-only - there is no `ragent backend` CLI subcommand.

2.1 Syntax

```
/backend                # open the switcher panel
/backend show           # alias of /backend
/backend list           # alias of /backend
/backend local          # switch to the local host backend
/backend docker         # switch to a Docker sandbox backend
/backend podman         # switch to a Podman sandbox backend
/backend remote         # switch to a remote backend
/backend <id>           # switch to a backend by its registered id
/backend help           # usage help
```

Form	Description
`/backend`	Open the switcher panel: the active backend and every registered backend's health.
`/backend show` / `/backend list`	Alias of `/backend`.
`/backend <kind>`	Switch directly to a backend *kind* (`local`, `docker`, `podman`, `remote`). Persists the bare kind label.

Form	Description
`/backend <id>`	Switch directly to a backend by its registered *id* (a `backends` entry, or an inline `execution_backend` descriptor).
`/backend help`	Print the usage table. Read-only; changes no state.

The selector is matched case-insensitively against a row's `id` **or** its `kind` (`switch_backend_by_selector`, `backend_panel.rs`).

2.2 The switcher panel

With no argument the command opens a **modal switcher panel** titled `Execution backend - active: <kind>`. Each row carries the backend's `id`, kind, health, and a secret-free connection summary; the active row carries an `[active]` marker.

Key	Action
`Up` / `Down`	Move the block cursor between rows.
`Enter`	Switch the active backend to the highlighted row.
`Esc`	Close the panel.

While the panel is open it owns the keyboard and the message input field stays locked. A row whose health is not `ok` is dimmed and **cannot** be selected: the switch is refused and the active backend is left unchanged, so the switcher can never leave the session on a backend that cannot run a tool. The rows are a snapshot taken when the panel opens (`BackendPanelState::build`); the panel does not re-probe health on each frame.

2.3 Health states

Health is computed from the config and the host, with a live probe only for `remote` entries (`initial_health` / `refresh_health` in `crates/ragent-agent/src/backend/registry.rs`).

Kind	`ok` when	`unavailable` when
`local`	always - the host is always available (FR-019)	never
`docker` / `podman`	the runtime binary resolves on `PATH` (FR-026)	the binary is missing (the detail names it)
`remote`	a base URL and a bearer key are configured (FR-030)	either is absent; a live `GET /health` probe refines the state (no key is sent)

2.4 Persistence

A successful switch writes `execution_backend: "<kind>"` into the loaded config file - the project-local `.ragent/ragent.json` when one was loaded, otherwise the global config (`persist_active_backend` / `backend_config_path`) - using an atomic temp-file-then-rename update. The per-turn config cache is then invalidated, so the change takes effect on the **next** turn. After a switch the status bar shows `Backend:<kind>` in cyan.

Because the persistence is by **kind label**, a descriptor-free switch such as `/backend docker` writes the bare label. To carry an image, workspace, URL, or key, declare a full descriptor in the `backends` array (or as an inline `execution_backend` object) and select it with `/backend <id>`.

2.5 Output

- A switch appends a `From: /backend` message bubble and sets the status to `backend: <kind>`.
- A refusal appends a `[warn]` message explaining why and sets the status to `backend: unavailable` / `backend: unknown`.
- An unregistered id is refused with a pointer to `/backend`.
- No permission prompt is raised and no LLM call is made.

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3. How the Switchable Backend Works

All of the machinery lives in `crates/ragent-agent/src/backend/`.

File	Responsibility
`mod.rs`	The `ExecutionBackend` trait, `BackendError`, the adapters, and resolution/caching helpers.
`registry.rs`	The durable backend registry (FR-004) and remote registration (FR-030).
`detection.rs`	Container-runtime detection on `PATH`, with `podman` as the default (FR-026).
`container.rs`	The `docker`/`podman` sandbox backend (FR-020, FR-031, FR-035).
`remote.rs`	The remote REST+SSE backend (FR-021, FR-034).
`secrets.rs`	Spawn-time credential resolution and injection (FR-010).

3.1 The `ExecutionBackend` trait

```rust
pub trait ExecutionBackend: Send + Sync {
    fn kind(&self) -> ExecutionBackendKind;
    async fn execute_tool(&self, call: BackendToolCall<'_>) -> anyhow::Result<ToolOutput>;
}
```

`BackendToolCall` carries the `Tool` handle, its JSON input, and the `ToolContext` (working directory, config, storage). Four implementations ship:

- `LocalBackend` - runs the tool on the host, exactly as before.
- `ContainerBackend` - provisions a container and dispatches into it.
- `RemoteBackend` - relays the turn to a second server.
- `PendingBackend` - a placeholder that fails every call, so a non-local kind with no adapter fails rather than silently running on the host (FR-031).

3.2 Resolution on every dispatch

Each tool dispatch resolves the backend from the session's live config (`dispatch_tool_leaf` in `crates/ragent-agent/src/session/processor.rs` calls `resolve_backend_with_secrets`):

```rust
pub fn resolve_backend_with_secrets(
    config: &Config,
```

```
    working_dir: &std::path::Path,
    storage: Option<Arc<crate::storage::Storage>>,
) -> Arc<dyn ExecutionBackend>
```

- `local` -> `local_backend()`.
- `docker` / `podman` -> a `ContainerBackend` built from the effective descriptor, wired to the encrypted credential store.
- `remote` -> a `RemoteBackend`.
- An absent or unknown `execution_backend` key resolves to **`local`** (`Config::effective_execution_backend`, FR-019); a typo is flagged by `Config::has_unknown_execution_backend` rather than misresolving onto a more capable backend.

Resolved backends are cached per `(kind, descriptor, working dir, credential set)` so a session reuses one instance.

3.3 The backend registry

`BackendRegistry::from_config` builds the read model from the config (FR-004):

1. The built-in `local` entry, id `local`, display name `Local (host)`, always present and healthy (FR-019).
2. One entry per `backends` descriptor, deduplicated by id.
3. The entry the `execution_backend` key resolves to (so an inline descriptor is visible).
4. A synthesised entry for a bare kind label with no matching descriptor (for example `execution_backend: "podman"`), so the active backend is always visible.

Each entry exposes a `ConnectionDescriptor` that carries credential **names** and presence flags only - never a value (FR-010, FR-035). For a container the summary is of the form `podman container=<name> image=<img> workspace=/projects volume=<vol>`.

3.4 The per-session override

Automation runs (spec `openhands` T-016, FR-033) need to confine a run to a specific backend regardless of the ambient config. This is done with a process-global map in `crates/ragent-agent/src/automation/mod.rs`:

```
pub fn register_session_backend(session_id: &str, descriptor: BackendConfig);
pub fn clear_session_backend(session_id: &str);
pub fn session_backend_override(session_id: &str) -> Option<BackendConfig>;
pub fn apply_session_backend_override(config: &Arc<Config>, session_id: &str) -> Arc<Config>;
```

`apply_session_backend_override` is applied once per turn in `crates/ragent-agent/src/session/loop_steps.rs` (before hooks are parsed), so the whole turn - and every tool dispatch in it - is confined to the run's configured backend. A run with no override returns the shared `Arc` unchanged, so the common path stays allocation-free; a run with an override gets a modified clone whose `execution_backend` names the selected descriptor. A relay failure therefore never silently re-runs the turn on the host (FR-031).

3.5 The container backend

`ContainerBackend` (`container.rs`) answers the "where" half of a tool invocation:

- **Provisioning (FR-009, FR-020).** On the first dispatch it starts **one long-lived container** for the session:

```
<runtime> run -d --name ragent-sandbox-<hash> \
           -v ragent-sandbox-vol-<hash>:/projects -w /projects \
           -e NAME=VALUE ... \
           <image> sleep infinity
```

The image must provide a minimal POSIX shell userland (`sh`, `sed`, `Grep`, `find`, `base64`); the container command is `sleep infinity`, so the sandbox stays up for the whole session and each tool call is one cheap `exec`. The container is created before the first tool call, so the project is made available to it before any dispatch (FR-009). A default image is required - if the descriptor names none, provisioning fails.

- **Workspace isolation (FR-020, FR-035).** The workspace is a **container-named volume** (`ragent-sandbox-vol-<hash>`), never a host bind mount. A tool inside the container cannot read or write a host path - it is the strongest form of "shall not expose the host working directory". The container and volume names are derived deterministically from `(runtime, id, image, workspace, credential names, host working dir)`, so a restarted process reuses the same sandbox instead of leaking a new one, while a changed credential set gets a freshly-provisioned container (FR-010, FR-023).
- **Dispatch.** Only the shell and file/search tools are served:
  - `Bash` - validated by the **7-layer host validator** (`ragent_tools_core::bash::validate_shell_command`) before it is executed, so the sandbox can never be used to bypass the security model (FR-002, FR-037); it then runs as `exec <name> sh -lc '<command>'` with the workspace as the working directory.
  - `read`, `Write`, `create`, `append_file`, `rm`, `mkdir`, `list`, `Glob`, `Grep` - translated to a sandbox `exec` script (content is transported base64-encoded so it cannot be mangled by quoting) and confined to the workspace mount point.
  - **Every other tool is refused** - the container backend has no adapter for it and never falls back to host execution (FR-031). This means that while a container backend is active, tools outside that subset (for example `webfetch`, `codeindex_*`, `mf_*`) are refused for the turn rather than routed.
- **The `workspace` field.** A `BackendConfig.workspace` that is an **absolute path** overrides the container-side mount point; a relative value such as `"."` keeps the default `/projects`.

3.6 The remote backend

`RemoteBackend` (`remote.rs`) points at a second ragent server (usually on another host) and runs the session **against that server**. It does not execute tools locally; it relays the whole turn over REST+SSE and mirrors that server's event stream into the local event bus (FR-021):

```
POST <url>/sessions                            { "directory": "<dir>" }  -> { "id": ... }
POST <url>/sessions/<id>/messages             { "content": "<prompt>" } -> text/event-stream
POST <url>/sessions/<id>/permission/<req_id> { "decision": "allow" }
```

The `send_message` route both starts the turn and streams its events, so one request drives the whole turn (`relay_turn`). Frames are decoded into `RemoteUpdates` and re-published onto the **local** bus under the local session id (`crates/ragent-agent/src/session/remote_dispatch.rs`), so the TUI renders a remote turn exactly like a local one. The remote session is cached per local session id, so a follow-up message reuses the same remote session.

Unreachability (FR-034):

- A server that cannot be reached before the turn starts is a provisioning failure (`RemoteErrorKind::Unreachable`) and fails the turn with no local execution (FR-031).
- A stream that drops **part-way through** a turn is `RemoteErrorKind::Protocol` and likewise never re-runs the turn's tools locally (FR-034).
- The relay is bounded by `TURN_TIMEOUT` (15 minutes) and polls a cancellation flag, so an unresponsive server cannot hang the session.

A `remote` descriptor with no `url` fails at provisioning rather than falling back to host execution (`validate_remote_backend`).

3.7 Credential injection

A `docker`/`podman`/`remote` descriptor may list `credentials` - **names**, never values. At spawn time the backend resolves each name from the encrypted credential store (the same `provider_auth` table the `ragent auth` command writes to) and injects it (FR-010):

```
pub fn resolve_descriptor_secrets(
    descriptor: &BackendConfig,
    resolver: &dyn SecretResolver,
) -> anyhow::Result<Vec<ResolvedSecret>>;

pub fn container_env_args(secrets: &[ResolvedSecret]) -> Vec<String>; // ["-e", "NAME=VALUE", ...
```

- Values are injected as container environment variables (`-e NAME=VALUE`), never written into the image, the workspace, or `ragent.json` (FR-035).
- A name the store does not hold is a **hard provisioning failure** - the sandbox is never started with a required credential silently missing.
- Every resolved value is registered with the shared redaction registry, so any later rendering (a log line, a `/config` dump, an error echoed back by the runtime) masks it.
- A running container is reused only if it already carries every credential the descriptor names; a changed credential set re-provisions the container with a fresh environment while re-mounting the same named volume, so the conversation's files survive (FR-023) and secret rotation takes effect at the next spawn.

3.8 No fallback to host

The single rule that ties the subsystem together (FR-031): if a non-local backend fails to provision, the turn fails with a `BackendError` whose `BackendErrorKind` is `Provision` (or `Protocol` for a mid-turn remote drop). The provisioning error is **sticky** - it is replayed on every subsequent dispatch in that session - and the tool is **never** run on the host instead. The error text always names the backend and states plainly "the tool was not executed on the host".

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4. Configuration Reference

Two keys drive backend selection (see `config.md` section 7.36 for the full field table).

```
{
  // Terse form: select a kind by label.
  "execution_backend": "podman",
```

```
  // Registry form: declare backends and select one by id (or by kind label).
  "backends": [
    { "id": "sandbox", "name": "Podman sandbox", "kind": "podman",
      "image": "ghcr.io/yourorg/ragent-sandbox:latest", "workspace": "/projects",
      "credentials": ["EXAMPLE_API_KEY"] },
    { "id": "build-farm", "name": "Remote build host", "kind": "remote",
      "url": "http://10.0.0.5:9100", "api_key": "sk-ragent-remote" }
  ]
}
```

`execution_backend` accepts either a bare label (`local`, `docker`, `podman`, `remote`) or an inline descriptor object (`{ "id", "kind", "image", "workspace" }`). An object that names a registered `id` (with no `kind`) resolves its kind from the matching `backends` entry. An absent or unknown value resolves safely to `local` (FR-019).

`BackendConfig` fields:

Field	Type	Default	Description
`id`	`String`	`""`	Stable identifier used to select this backend.
`name`	`Option<String>`	None	Human-readable display name; defaults to `id`.
`kind`	`String`	`""`	Backend kind: `local`, `docker`, `podman`, or `remote`.
`image`	`Option<String>`	None	Container image reference (required for `docker`/`podman`).
`workspace`	`Option<String>`	None	Container-side workspace mount point. An absolute path overrides the default `/projects`; a relative value such as `"."` keeps the default.
`url`	`Option<String>`	None	Base URL of a remote server (`remote` kind).
`api_key`	`Option<String>`	None	Bearer key for a remote server (`remote` kind). Redacted in diagnostics. A literal `api_key` takes precedence over a named credential.
`credentials`	`Vec<String>`	`[]`	Names of encrypted-store credentials resolved at spawn time and injected as container environment variables (FR-010).

`podman` is the default container runtime when none is specified (FR-026).

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5. Worked Example: a Docker/Podman Sandbox

This walks through running a session inside a Podman sandbox, end to end. Docker works identically - substitute `docker` for `podman` and `"kind": "docker"`.

Step 1 - Confirm a runtime is present (FR-026)

```
podman --version          # or: docker --version
```

ragent probes `PATH` for the runtime binary (it never spawns the runtime just to check). If `podman` is missing, the `/backend` row for a Podman backend reads `unavailable` and cannot be selected. Install `podman` (or `docker`) or stay on `local`.

Estimated time: 2 minutes if Podman is already installed, 5-10 minutes to install it fresh.

Step 2 - Build a sandbox image

The image must provide a minimal POSIX shell userland. A small Debian-slim base is enough (`coreutils` supplies `base64`, `findutils` supplies `find`):

`Dockerfile`:

```
FROM debian:bookworm-slim
RUN apt-get update \
 && apt-get install -y --no-install-recommends \
      coreutils findutils grep sed ca-certificates \
 && rm -rf /var/lib/apt/lists/*
# The backend starts the container as `sleep infinity`, so no ENTRYPOINT/CMD
# is required; anything that keeps PID 1 alive works.
```

Build it with the runtime that will back the sandbox:

```
podman build -t ragent-sandbox:latest -f Dockerfile .
```

(The backend starts the container with `sleep infinity`, so the image does not need a long-running entrypoint.)

Step 3 - Store the credential the sandbox needs

A descriptor's `credentials` list names entries in the encrypted credential store - the same `provider_auth` table the `ragent auth` command writes to. Store the value under the name the descriptor will list (never put the value in `ragent.json`):

```
ragent auth EXAMPLE_API_KEY sk-example-0123456789
```

The value is stored encrypted and resolved by name **at spawn time**, then injected as the container environment variable `EXAMPLE_API_KEY`.

Step 4 - Declare the backend

Add the descriptor and select it. In the project-local `.ragent/ragent.json`:

```
{
  "execution_backend": "sandbox",
  "backends": [
    {
      "id": "sandbox",
      "name": "Podman sandbox",
      "kind": "podman",
      "image": "ragent-sandbox:latest",
```

```
      "workspace": "/projects",
      "credentials": ["EXAMPLE_API_KEY"]
    }
  ]
}
```

Notes:

- `workspace` is the **container-side** mount point. `"/projects"` is the default; a relative value such as `"."` also keeps the default.
- `credentials` carries **names only** - `EXAMPLE_API_KEY` here, matching the `ragent auth EXAMPLE_API_KEY ...` above.
- Omit `execution_backend` (or set it to `"local"`) to keep host execution.

Step 5 - Activate the backend

Start the TUI and check the switcher:

```
/backend
```

The panel lists `local` (active) and `sandbox` (kind `podman`, health `ok`). Move with `Up`/`Down` and press `Enter` on `sandbox` - or select it in one step:

```
/backend sandbox
```

Either way, ragent persists `execution_backend: "sandbox"` to the config file and the status bar now shows `Backend:podman`. The change applies to the **next** turn.

Step 6 - Run a turn and observe the sandbox

Ask the agent to inspect the workspace:

```
List the files in the workspace with `Bash: ls -la`.
```

What happens on the first tool call:

1. ragent provisions the sandbox container: `<runtime> run -d --name ragent-sandbox-<hash> -v ragent-sandbox-vol-<hash>:/projects -w /projects -e EXAMPLE_API_KEY=<value> ragent-sandbox:latest sleep infinity`.
2. The `ls -la` command is validated by the **7-layer bash validator** on the host as usual, then executed inside the container as `exec <name> sh -lc 'ls -la'`.

You will see that `/projects` starts **empty** - the host working directory is not mounted. A `read` or `ls` of a host path inside the sandbox finds nothing, and a file you create inside the sandbox does **not** appear on the host. That is FR-020/FR-035 working as designed.

> If the model tries a tool outside the sandbox subset (for example `webfetch` or `codeindex_search`), it is refused for that turn - the container backend has no adapter for it and will not run it on the host (FR-031).

Step 7 - Fail-without-fallback (prove FR-031)

Point the descriptor at an image that does not exist:

```
"image": "ragent-sandbox:nonexistent"
```

The next tool call fails with a `Provision` error naming the image and stating the tool was not executed on the host. Fixing the image and re-running the turn provisions cleanly; the same named volume is re-mounted, so any files the conversation created earlier are still there.

Step 8 - Rotate a credential

Change the stored value:

```
ragent auth EXAMPLE_API_KEY sk-example-rotated
```

Because the credential set is part of the sandbox's deterministic identity, the next spawn **replaces** the container with a fresh environment carrying the new value while re-mounting the same volume (FR-010, FR-023).

Step 9 - Return to the host and clean up

```
/backend local
```

The status bar returns to `Backend:local`. To reclaim disk used by the sandbox:

```
podman rm -f $(podman ps -a --filter name=ragent-sandbox --format '{{.Names}}')
podman volume rm $(podman volume ls --filter name=ragent-sandbox --format '{{.Name}}')
```

---

6. Troubleshooting

Symptom	Cause	Fix
`/backend` row for the sandbox is `unavailable` and cannot be selected	the runtime binary is not on `PATH` (FR-026)	install `podman`/`docker`, or select `local`.
Turn fails with "container runtime … was not found on PATH"	the descriptor names a runtime that is absent	install it, or drop `image`/switch the descriptor's `kind`.
Turn fails with "has no container image configured"	`image` is missing on the descriptor	set `image`.
Turn fails with "failed to start the … sandbox container from image …"	the image cannot be pulled/started	build or pull the image; the runtime's own error is echoed (with secrets redacted).
Turn fails with "requires the credential … which is not present"	the `credentials` name is not in the store	store it with `ragent auth <NAME> <VALUE>` (FR-010).
A tool is refused with "no adapter"	the tool is outside the sandbox's shell/file/search subset	use a tool in the subset, or switch to `local` (FR-031).
`/backend <id>` reports "No registered backend matches"	the id is not declared in `backends`	add a descriptor, or use a kind label.

7. See Also

- `slashcommands/backend.md` - the `/backend` command quick reference.
- `config.md` section 7.36 - the `execution_backend` and `backends` config keys.
- `permissions.md` - the static permission pipeline and the 7-layer bash model that run before any backend dispatch.
- `llmsecurity.md` - the optional LLM security analyzer, a tightening-only layer over the same permission system.
- `specs/openhands/SPEC.md` - FR-001, FR-004, FR-008, FR-009, FR-010, FR-019, FR-020, FR-021, FR-023, FR-026, FR-030, FR-031, FR-033, FR-034, FR-035.